

Laplace Transform

$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

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MATHEMATICS FOR INFORMATION TECHNOLOGY

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Outline



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

- Matrices
 - The Inverse of a Matrix
- Determinants
 - The Determinant of a Matrix
 - Applications of Determinants
- Applications of Matrix Operations
 - Cryptography

The Inverse of a Matrix

A decorative image in the top right corner showing a blurred snippet of a document with the Laplace Transform formula: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.

An $n \times n$ matrix A is **invertible** (or **nonsingular**) if there exists an $n \times n$ matrix B such that

$$AB = BA = I_n$$

where I_n is the identity matrix of order n . The matrix B is called the (multiplicative) **inverse** of A . A matrix that does not have an inverse is called **noninvertible** (or **singular**).

Finding the Inverse of a Matrix

Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}.$$

SOLUTION

To find the inverse of A , try to solve the matrix equation $AX = I$ for X .

$$\begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x_{11} + 4x_{21} & x_{12} + 4x_{22} \\ -x_{11} - 3x_{21} & -x_{12} - 3x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Now, by equating corresponding entries, you obtain the two systems of linear equations shown below.

$$\begin{array}{ll} x_{11} + 4x_{21} = 1 & x_{12} + 4x_{22} = 0 \\ -x_{11} - 3x_{21} = 0 & -x_{12} - 3x_{22} = 1 \end{array}$$

Properties of Inverse Matrices



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

If A is an invertible matrix, k is a positive integer, and c is a scalar not equal to zero, then A^{-1} , A^k , cA , and A^T are invertible and the following are true.

1. $(A^{-1})^{-1} = A$
2. $(A^k)^{-1} = A^{-1}A^{-1} \cdot \cdot \cdot A^{-1} = (A^{-1})^k$
3. $(cA)^{-1} = \frac{1}{c}A^{-1}, c \neq 0$
4. $(A^T)^{-1} = (A^{-1})^T$

If A and B are invertible matrices of size n , then AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

The Determinant of a Matrix

- Every *square* matrix can be associated with a real number called its **determinant**.
- Historically, the use of determinants arose from the recognition of special patterns that occur in the solutions of systems of linear equations.

Definition of the Determinant of a 2×2 Matrix

The **determinant** of the matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

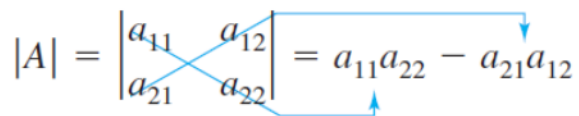
is given by

$$\det(A) = |A| = a_{11}a_{22} - a_{21}a_{12}.$$

REMARK: In this text, $\det(A)$ and $|A|$ are used interchangeably to represent the determinant of a matrix. Vertical bars are also used to denote the absolute value of a real number; the context will show which use is intended. Furthermore, it is common practice to delete the matrix brackets and write

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \quad \text{instead of} \quad \left| \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \right|.$$

A convenient method for remembering the formula for the determinant of a 2×2 matrix is shown in the diagram below.

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$


The determinant is the difference of the products of the two diagonals of the matrix. Note that the order is important, as demonstrated above.

The Determinant of a Matrix of Order 2

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Find the determinant of each matrix.

(a) $A = \begin{bmatrix} 2 & -3 \\ 1 & 2 \end{bmatrix}$ (b) $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ (c) $C = \begin{bmatrix} 0 & 3 \\ 2 & 4 \end{bmatrix}$

SOLUTION (a) $|A| = \begin{vmatrix} 2 & -3 \\ 1 & 2 \end{vmatrix} = 2(2) - 1(-3) = 4 + 3 = 7$

(b) $|B| = \begin{vmatrix} 2 & 1 \\ 4 & 2 \end{vmatrix} = 2(2) - 4(1) = 4 - 4 = 0$

(c) $|C| = \begin{vmatrix} 0 & 3 \\ 2 & 4 \end{vmatrix} = 0(4) - 2(3) = 0 - 6 = -6$

Definitions of Minors and Cofactors of a Matrix

$$\text{Laplace-Trans} \\ f(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

If A is a square matrix, then the **minor** M_{ij} of the element a_{ij} is the determinant of the matrix obtained by deleting the i th row and j th column of A . The **cofactor** C_{ij} is given by

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

For example, if A is a 3×3 matrix, then the minors and cofactors of a_{21} and a_{22} are as shown in the diagram below.

Minor of a_{21}

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

Delete row 2 and column 1.

Cofactor of a_{21}

$$\begin{aligned} C_{21} &= (-1)^{2+1} M_{21} \\ &= -M_{21} \end{aligned}$$

Minor of a_{22}

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad M_{22} = \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$$

Delete row 2 and column 2.

Cofactor of a_{22}

$$\begin{aligned} C_{22} &= (-1)^{2+2} M_{22} \\ &= M_{22} \end{aligned}$$

Definitions of Minors and Cofactors of a Matrix (cont.)

As you can see, the minors and cofactors of a matrix can differ only in sign. To obtain the cofactors of a matrix, first find the minors and then apply the checkerboard pattern of +’s and -’s shown below.

Sign Pattern for Cofactors

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

3×3 matrix

$$\begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix}$$

4×4 matrix

$$\begin{bmatrix} + & - & + & - & + & \dots \\ - & + & - & + & - & \dots \\ + & - & + & - & + & \dots \\ - & + & - & + & - & \dots \\ + & - & + & - & + & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

$n \times n$ matrix

Note that *odd* positions (where $i + j$ is odd) have negative signs, and even positions (where $i + j$ is even) have positive signs.

Find the Minors and Cofactors of a Matrix



Laplace-Transform
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

Find all the minors and cofactors of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

SOLUTION

To find the minor M_{11} , delete the first row and first column of A and evaluate the determinant of the resulting matrix.

$$\begin{bmatrix} \textcircled{0} & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}, \quad M_{11} = \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} = -1(1) - 0(2) = -1$$

Similarly, to find M_{12} , delete the first row and second column.

$$\begin{bmatrix} 0 & \textcircled{2} & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}, \quad M_{12} = \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix} = 3(1) - 4(2) = -5$$

Continuing this pattern, you obtain

$$\begin{array}{lll} M_{11} = -1 & M_{12} = -5 & M_{13} = 4 \\ M_{21} = 2 & M_{22} = -4 & M_{23} = -8 \\ M_{31} = 5 & M_{32} = -3 & M_{33} = -6. \end{array}$$

Now, to find the cofactors, combine the checkerboard pattern of signs with these minors to obtain

$$\begin{array}{lll} C_{11} = -1 & C_{12} = 5 & C_{13} = 4 \\ C_{21} = -2 & C_{22} = -4 & C_{23} = 8 \\ C_{31} = 5 & C_{32} = 3 & C_{33} = -6. \end{array}$$

Definition of the Determinant of a Matrix

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If A is a square matrix (of order 2 or greater), then the determinant of A is the sum of the entries in the first row of A multiplied by their cofactors. That is,

$$\det(A) = |A| = \sum_{j=1}^n a_{1j}C_{1j} = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}.$$

The Determinant of a Matrix of Order 3



Laplace-Transform
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

Find the determinant of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

SOLUTION This matrix is the same as the one in Example 2. There you found the cofactors of the entries in the first row to be

$$C_{11} = -1, \quad C_{12} = 5, \quad C_{13} = 4.$$

By the definition of a determinant, you have

$$\begin{aligned} |A| &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} && \text{First row expansion} \\ &= 0(-1) + 2(5) + 1(4) = 14. \end{aligned}$$

Although the determinant is defined as an expansion by the cofactors in the first row, it can be shown that the determinant can be evaluated by expanding by *any* row or column. For instance, you could expand the 3×3 matrix in Example 3 by the second row to obtain

$$\begin{aligned} |A| &= a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23} && \text{Second row expansion} \\ &= 3(-2) + (-1)(-4) + 2(8) = 14 \end{aligned}$$

or by the first column to obtain

$$\begin{aligned} |A| &= a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31} && \text{First column expansion} \\ &= 0(-1) + 3(-2) + 4(5) = 14. \end{aligned}$$

Expansion by Cofactors



Laplace-Transform
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

Let A be a square matrix of order n . Then the determinant of A is given by

$$\det(A) = |A| = \sum_{j=1}^n a_{ij}C_{ij} = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

or

$$\det(A) = |A| = \sum_{i=1}^n a_{ij}C_{ij} = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}.$$

The Determinant of a Matrix of Order 4



Laplace-Transform
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

Find the determinant of

$$A = \begin{bmatrix} 1 & -2 & 3 & 0 \\ -1 & 1 & 0 & 2 \\ 0 & 2 & 0 & 3 \\ 3 & 4 & 0 & -2 \end{bmatrix}.$$

SOLUTION

By inspecting this matrix, you can see that three of the entries in the third column are zeros. You can eliminate some of the work in the expansion by using the third column.

$$|A| = 3(C_{13}) + 0(C_{23}) + 0(C_{33}) + 0(C_{43})$$

Because C_{23} , C_{33} , and C_{43} have zero coefficients, you need only find the cofactor C_{13} . To do this, delete the first row and third column of A and evaluate the determinant of the resulting matrix.

$$C_{13} = (-1)^{1+3} \begin{vmatrix} -1 & 1 & 2 \\ 0 & 2 & 3 \\ 3 & 4 & -2 \end{vmatrix} = \begin{vmatrix} -1 & 1 & 2 \\ 0 & 2 & 3 \\ 3 & 4 & -2 \end{vmatrix}$$

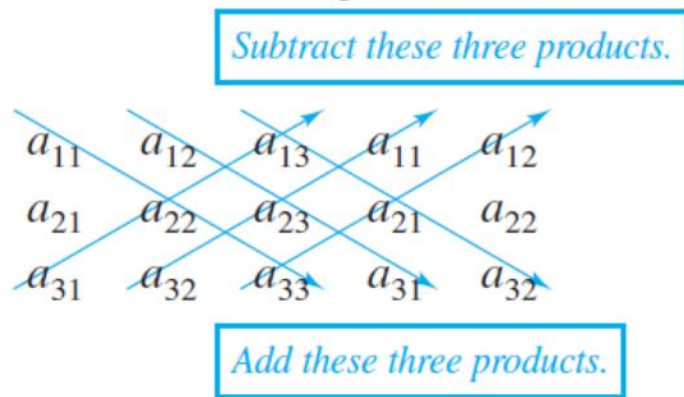
Expanding by cofactors in the second row yields

$$\begin{aligned} C_{13} &= (0)(-1)^{2+1} \begin{vmatrix} 1 & 2 \\ 4 & -2 \end{vmatrix} + (2)(-1)^{2+2} \begin{vmatrix} -1 & 2 \\ 3 & -2 \end{vmatrix} + (3)(-1)^{2+3} \begin{vmatrix} -1 & 1 \\ 3 & 4 \end{vmatrix} \\ &= 0 + 2(1)(-4) + 3(-1)(-7) = 13. \end{aligned}$$

You obtain $|A| = 3(13) = 39$.

Definition of the Determinant of a Matrix (cont.)

There is an alternative method commonly used for evaluating the determinant of a 3×3 matrix A . To apply this method, copy the first and second columns of A to form fourth and fifth columns. The determinant of A is then obtained by adding (or subtracting) the products of the six diagonals, as shown in the following diagram.



Try confirming that the determinant of A is

$$|A| = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}.$$

The Determinant of a Matrix of Order 3

$$\text{Laplace-Trans} \\ F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

Find the determinant of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & -4 & 1 \end{bmatrix}.$$

SOLUTION

Begin by recopying the first two columns and then computing the six diagonal products as follows.

0	2	1	0	2
3	-1	2	3	-1
4	-4	1	4	-4

-4 0 6
0 16 -12

← Subtract these products.

← Add these products.

Now, by adding the lower three products and subtracting the upper three products, you can find the determinant of A to be $|A| = 0 + 16 + (-12) - (-4) - 0 - 6 = 2$.

Conditions That Yield a Zero Determinant

Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

If A is a square matrix and any one of the following conditions is true, then $\det(A) = 0$.

1. An entire row (or an entire column) consists of zeros.
2. Two rows (or columns) are equal.
3. One row (or column) is a multiple of another row (or column).

$$\begin{vmatrix} 0 & 0 & 0 \\ 2 & 4 & -5 \\ 3 & -5 & 2 \end{vmatrix} = 0,$$

*The first row
has all zeros.*

$$\begin{vmatrix} 1 & -2 & 4 \\ 0 & 1 & 2 \\ 1 & -2 & 4 \end{vmatrix} = 0,$$

*The first and third
rows are the same.*

$$\begin{vmatrix} 1 & 2 & -3 \\ 2 & -1 & -6 \\ -2 & 0 & 6 \end{vmatrix} = 0.$$

*The third column is a
multiple of the first column.*

Properties of Determinants

If A and B are square matrices of order n , then

$$\det(AB) = \det(A) \det(B).$$

If A is an $n \times n$ matrix and c is a scalar, then the determinant of cA is given by

$$\det(cA) = c^n \det(A).$$

A square matrix A is invertible (nonsingular) if and only if

$$\det(A) \neq 0.$$

If A is invertible, then

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

If A is a square matrix, then

$$\det(A) = \det(A^T).$$

The Adjoint of a Matrix

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

- If A is a square matrix, then the **matrix of cofactors** of A has the form

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}.$$

- The transpose of this matrix is called the **adjoint** of A and is denoted by $\text{adj}(A)$. That is,

$$\text{adj}(A) = \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}.$$

Finding the Adjoint of a Square Matrix

Laplace-Transform
 $F(p) = \int_0^\infty e^{-pt} f(t) dt$

Find the adjoint of

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}.$$

SOLUTION The cofactor C_{11} is given by

$$\begin{bmatrix} \boxed{-1} & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix} \rightarrow C_{11} = (-1)^2 \begin{vmatrix} -2 & 1 \\ 0 & -2 \end{vmatrix} = 4.$$

Continuing this process produces the following matrix of cofactors of A .

$$\begin{bmatrix} \begin{vmatrix} -2 & 1 \\ 0 & -2 \end{vmatrix} & -\begin{vmatrix} 0 & 1 \\ 1 & -2 \end{vmatrix} & \begin{vmatrix} 0 & -2 \\ 1 & 0 \end{vmatrix} \\ -\begin{vmatrix} 3 & 2 \\ 0 & -2 \end{vmatrix} & \begin{vmatrix} -1 & 2 \\ 1 & -2 \end{vmatrix} & -\begin{vmatrix} -1 & 3 \\ 1 & 0 \end{vmatrix} \\ \begin{vmatrix} 3 & 2 \\ -2 & 1 \end{vmatrix} & -\begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} & \begin{vmatrix} -1 & 3 \\ 0 & -2 \end{vmatrix} \end{bmatrix} = \begin{bmatrix} 4 & 1 & 2 \\ 6 & 0 & 3 \\ 7 & 1 & 2 \end{bmatrix}$$

The transpose of this matrix is the adjoint of A . That is,

$$\text{adj}(A) = \begin{bmatrix} 4 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix}.$$

The Inverse of a Matrix Given by Its Adjoint

A decorative image in the top right corner showing a portion of a document with the text 'Laplace-Trans' and the formula $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

If A is an $n \times n$ invertible matrix, then

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

Using the Adjoint of a Matrix to Find Its Inverse

Use the adjoint of

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}$$

to find A^{-1} .

SOLUTION The determinant of this matrix is 3. Using the adjoint of A (found in Example 1), you can find the inverse of A to be

$$A^{-1} = \frac{1}{|A|} \text{adj}(A) = \frac{1}{3} \begin{bmatrix} 4 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix} = \begin{bmatrix} \frac{4}{3} & 2 & \frac{7}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & 1 & \frac{2}{3} \end{bmatrix}.$$

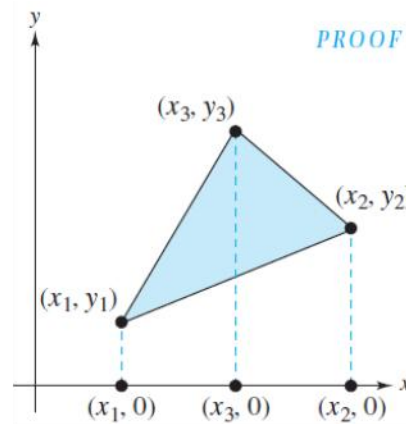
You can check to see that this matrix is the inverse of A by multiplying to obtain

$$AA^{-1} = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix} \begin{bmatrix} \frac{4}{3} & 2 & \frac{7}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & 1 & \frac{2}{3} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Applications of Determinants

- Area, Volume, and Equations of Lines and Planes
- Two-Point Form of the Equation of a Line
- Volume of a Tetrahedron

Area, Volume, and Equations of Lines and Planes



The area of the triangle whose vertices are (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) is given by

$$\text{Area} = \pm \frac{1}{2} \det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix},$$

where the sign $(\pm)$ is chosen to yield a positive area.

Finding the Area of a Triangle

Find the area of the triangle whose vertices are $(1, 0)$, $(2, 2)$, and $(4, 3)$.

SOLUTION

It is not necessary to know the relative positions of the three vertices. Simply evaluate the determinant

$$\frac{1}{2} \begin{vmatrix} 1 & 0 & 1 \\ 2 & 2 & 1 \\ 4 & 3 & 1 \end{vmatrix} = -\frac{3}{2}$$

and conclude that the area of the triangle is $\frac{3}{2}$.

Two-Point Form of the Equation of a Line



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

An equation of the line passing through the distinct points (x_1, y_1) and (x_2, y_2) is given by

$$\det \begin{bmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{bmatrix} = 0.$$

Finding an Equation of the Line Passing Through Two Points

Find an equation of the line passing through the points $(2, 4)$ and $(-1, 3)$.

SOLUTION

Applying the determinant formula for the equation of a line passing through two points produces

$$\begin{vmatrix} x & y & 1 \\ 2 & 4 & 1 \\ -1 & 3 & 1 \end{vmatrix} = 0.$$

To evaluate this determinant, expand by cofactors along the top row to obtain

$$x \begin{vmatrix} 4 & 1 \\ 3 & 1 \end{vmatrix} - y \begin{vmatrix} 2 & 1 \\ -1 & 1 \end{vmatrix} + 1 \begin{vmatrix} 2 & 4 \\ -1 & 3 \end{vmatrix} = 0$$
$$x - 3y + 10 = 0.$$

An equation of the line is $x - 3y = -10$.

Volume of a Tetrahedron

Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

- เตตระฮีดรอน ตรงกับภาษาอังกฤษว่า Tetrahedron (อ่านว่า tet-tra-he-dron) หมายถึงโครงสร้างปริมาตรฐานสามเหลี่ยม ซึ่งประกอบด้วย ด้าน (face) สามเหลี่ยม 4 ด้าน มุม (vertice) 4 มุม ขอบ (edge) 6 ขอบ เตตระฮีดรอนเป็นรูปทรงที่จัดอยู่ในประเภท พลาโตนิก (platonic)

The volume of the tetrahedron whose vertices are (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) , and (x_4, y_4, z_4) is given by

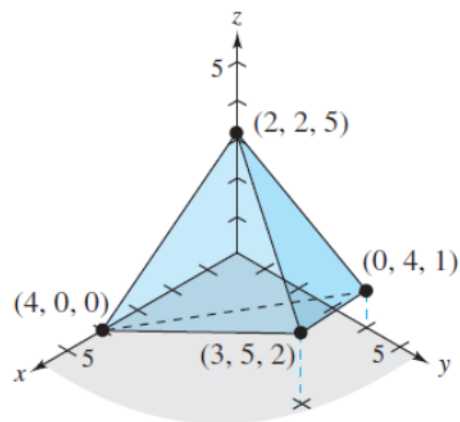
$$\text{Volume} = \pm \frac{1}{6} \det \begin{bmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{bmatrix},$$

where the sign $(\pm)$ is chosen to yield a positive volume.

Finding the Volume of a Tetrahedron

$$\text{Laplace-Trans} \\ F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

Find the volume of the tetrahedron whose vertices are $(0, 4, 1)$, $(4, 0, 0)$, $(3, 5, 2)$, and $(2, 2, 5)$



SOLUTION Using the determinant formula for volume produces

$$\frac{1}{6} \begin{vmatrix} 0 & 4 & 1 & 1 \\ 4 & 0 & 0 & 1 \\ 3 & 5 & 2 & 1 \\ 2 & 2 & 5 & 1 \end{vmatrix} = \frac{1}{6}(-72) = -12.$$

The volume of the tetrahedron is 12.

Applications of Matrix Operations

- Cryptography

A decorative header image showing a close-up of a document with the Laplace Transform formula. The text "Laplace-Trans" is visible at the top, followed by the equation $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

Cryptography

A decorative image in the top right corner showing a mathematical formula for the Laplace Transform: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$. The text "Laplace-Trans" is visible above the formula.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

- A **cryptogram** is a message written according to a secret code (the Greek word *kryptos* means “hidden”).
- This section describes a method of using matrix multiplication to **encode** and **decode** messages.

$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$


Cryptography (cont.)



Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

- Begin by assigning a number to each letter in the alphabet (with 0 assigned to a blank space), as follows.

0 = _	14 = N
1 = A	15 = O
2 = B	16 = P
3 = C	17 = Q
4 = D	18 = R
5 = E	19 = S
6 = F	20 = T
7 = G	21 = U
8 = H	22 = V
9 = I	23 = W
10 = J	24 = X
11 = K	25 = Y
12 = L	26 = Z
13 = M	

- Then the message is converted to numbers and partitioned into **uncoded row matrices**, each having n entries.

Forming Uncoded Row Matrices



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Write the uncoded row matrices of size 1×3 for the message MEET ME MONDAY.

SOLUTION Partitioning the message (including blank spaces, but ignoring punctuation) into groups of three produces the following uncoded row matrices.

$$\begin{array}{ccccccc} [13 & 5 & 5] & [20 & 0 & 13] & [5 & 0 & 13] & [15 & 14 & 4] & [1 & 25 & 0] \\ M & E & E & T & _ & M & E & _ & M & O & N & D & A & Y & _ \end{array}$$

Note that a blank space is used to fill out the last uncoded row matrix.

Cryptography (cont.)



Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

- To **encode** a message, choose an $n \times n$ invertible matrix A and multiply the uncoded row matrices (on the right) by A to obtain **coded row matrices**.

Encoding a Message



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

Use the matrix

$$A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

to encode the message MEET ME MONDAY.

SOLUTION The coded row matrices are obtained by multiplying each of the uncoded row matrices found in Example 4 by the matrix A , as follows.

Uncoded Row Matrix	Encoding Matrix A	Coded Row Matrix
$[13 \quad 5 \quad 5]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$= [13 \quad -26 \quad 21]$
$[20 \quad 0 \quad 13]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$= [33 \quad -53 \quad -12]$
$[5 \quad 0 \quad 13]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$= [18 \quad -23 \quad -42]$
$[15 \quad 14 \quad 4]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$= [5 \quad -20 \quad 56]$
$[1 \quad 25 \quad 0]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$= [-24 \quad 23 \quad 77]$

The sequence of coded row matrices is

$$[13 \quad -26 \quad 21][33 \quad -53 \quad -12][18 \quad -23 \quad -42][5 \quad -20 \quad 56][-24 \quad 23 \quad 77].$$

Finally, removing the brackets produces the cryptogram below.

$$13 \quad -26 \quad 21 \quad 33 \quad -53 \quad -12 \quad 18 \quad -23 \quad -42 \quad 5 \quad -20 \quad 56 \quad -24 \quad 23 \quad 77$$

Cryptography (cont.)

A decorative image in the top right corner showing a blurred snippet of a document with the Laplace Transform formula: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.

- For those who do not know the matrix A , decoding the cryptogram found in the previous example is difficult.
- But for an authorized receiver who knows the matrix A , decoding is simple.
- The receiver need only multiply the coded row matrices by A^{-1} to retrieve the uncoded row matrices.
- In other words, if

$$X = [x_1 \ x_2 \ \cdots \ x_n]$$

is an uncoded $1 \times n$ matrix, then $Y = XA$ is the corresponding encoded matrix.

- The receiver of the encoded matrix can decode Y by multiplying on the right by A^{-1} to obtain

$$YA^{-1} = (XA)A^{-1} = X.$$

Decoding a Message

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Use the inverse of the matrix

$$A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

to decode the cryptogram

$$13 \quad -26 \quad 21 \quad 33 \quad -53 \quad -12 \quad 18 \quad -23 \quad -42 \quad 5 \quad -20 \quad 56 \quad -24 \quad 23 \quad 77.$$

SOLUTION Begin by using Gauss-Jordan elimination to find A^{-1} .

$$\begin{array}{c} [A : I] \\ \left[\begin{array}{ccc|ccc} 1 & -2 & 2 & 1 & 0 & 0 \\ -1 & 1 & 3 & 0 & 1 & 0 \\ 1 & -1 & -4 & 0 & 0 & 1 \end{array} \right] \end{array} \rightarrow \begin{array}{c} [I : A^{-1}] \\ \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -10 & -8 \\ 0 & 1 & 0 & -1 & -6 & -5 \\ 0 & 0 & 1 & 0 & -1 & -1 \end{array} \right] \end{array}$$

Now, to decode the message, partition the message into groups of three to form the coded row matrices

$$[13 \quad -26 \quad 21][33 \quad -53 \quad -12][18 \quad -23 \quad -42][5 \quad -20 \quad 56][-24 \quad 23 \quad 77].$$

Decoding a Message (cont.)

To obtain the decoded row matrices, multiply each coded row matrix by A^{-1} (on the right).

<i>Coded Row Matrix</i>	<i>Decoding Matrix A^{-1}</i>	<i>Decoded Row Matrix</i>
$[13 \ -26 \ 21]$	$\begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix}$	$= [13 \ 5 \ 5]$
$[33 \ -53 \ -12]$	$\begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix}$	$= [20 \ 0 \ 13]$
$[18 \ -23 \ -42]$	$\begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix}$	$= [5 \ 0 \ 13]$
$[5 \ -20 \ 56]$	$\begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix}$	$= [15 \ 14 \ 4]$
$[-24 \ 23 \ 77]$	$\begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix}$	$= [1 \ 25 \ 0]$

The sequence of decoded row matrices is

$$[13 \ 5 \ 5][20 \ 0 \ 13][5 \ 0 \ 13][15 \ 14 \ 4][1 \ 25 \ 0]$$

and the message is

13 5 5 20 0 13 5 0 13 15 14 4 1 25 0.
M E E T _ M E _ M O N D A Y _

Q & A

A close-up, slightly blurred image of a document page. The text 'Laplace-Trans' is visible at the top. Below it, the Laplace transform formula is written:
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

with the text 'with the condition' partially visible below the formula.